

Exploring Predictors of Students' Mathematics Self-Efficacy Using glmLasso: Evidence From South Korea and Singapore PISA 2022 Data

SAGE Open
Vol. 16(3): 1–17
© The Author(s) 2026
DOI: 10.1177/21582440261475451
journals.sagepub.com/home/sgo
 SAGE

Myoungsung Lee¹ , Seongmin Park¹ , and Sehee Hong¹

Abstract

As artificial intelligence, science, and technology advance, mathematical self-efficacy for 21st-century reasoning tasks has become increasingly important. This study aimed to explore student- and school-level predictors of mathematics self-efficacy among 15-year-olds in South Korea and Singapore—two countries that rank high in mathematics achievement but differ in student self-efficacy as measured by the Programme for International Student Assessment (PISA) 2022. This study focused specifically on the newly released PISA 2022 self-efficacy scale, which assesses students' mathematical reasoning and 21st-century mathematical tasks. To identify potential predictors of mathematics self-efficacy, we applied a variable selection approach using glmLasso, a machine learning method that accounts for multilevel structure. The selected predictors were subsequently tested for statistical significance using multilevel linear models. A total of 108 predictors were considered, including 52 student- and 56 school-level variables. The results identified 14 student-level predictors associated with mathematics self-efficacy in both countries. Regarding country-specific predictors, seven student-level predictors were identified in South Korea, whereas in Singapore, two student- and two school-level predictors were identified. This study examined common and country-specific predictors of mathematics self-efficacy across the two countries and provided implications for its improvement.

Plain Language Summary

This study examined mathematics self-efficacy, which refers to students' belief in their ability to succeed in mathematics. The study focused on 15-year-old students in South Korea and Singapore, two countries that both perform highly in mathematics but differ in their levels of mathematics self-efficacy. We used data from the Programme for International Student Assessment (PISA) 2022. In this study, mathematics self-efficacy was measured in relation to reasoning and 21st-century mathematics tasks, which were included in the newly released PISA items. To identify which factors predicted mathematics self-efficacy, we analyzed 108 candidate variables. These included 52 student-level variables, such as background, attitudes, and behaviors, and 56 school-level variables, such as resources and school environment. To select the most important variables, we used a machine learning method called glmLasso, which is used to identify predictors in large and complex datasets. After this step, we tested whether the selected variables were statistically significant using multilevel linear models. The analysis showed that in South Korea, 21 student-level predictors were identified, while in Singapore, 16 student-level and two school-level predictors were identified. Of these, 14 student-level predictors were significant in both countries, showing common patterns. At the same time, seven predictors were unique to South Korea, while two student-level and two school-level predictors were unique to Singapore. This study contributes by identifying both common and country-specific

¹Department of Education, Korea University, Seoul, Republic of Korea

Corresponding author:

Sehee Hong, Department of Education, Korea University, 145, Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea.
Email: seeehong@korea.ac.kr



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

predictors of mathematics self-efficacy. These results offer new insights into the factors associated with mathematics self-efficacy.

Keywords

mathematics self-efficacy, PISA 2022, South Korea, Singapore, glmLasso

Received: 2 September 2025; Revised: 6 April 2026; Accepted: 16 July 2026

Introduction

With the rapid growth of big data and artificial intelligence, mathematics has become a key subject in 21st-century education. As “the pace of mathematization of the world has been accelerating” (Davis, 1988, p. 139), complex challenges in society and everyday life require students to think mathematically and develop the skills to apply mathematics in real-world contexts (Organisation for Economic Co-operation and Development [OECD], 2023). With the development of technology that drives the world toward a more data-centered society, a strong foundation in mathematics is essential for adapting to ongoing changes and enhancing national competitiveness. Reflecting this importance, countries worldwide have continually sought to reform their mathematics curricula and evaluation systems to effectively improve student achievement.

Against this backdrop, the Programme for International Student Assessment (PISA), one of the major international large-scale assessments conducted by the OECD (2024c), triennially tracks the mathematical achievement of 15-year-old students worldwide while also examining student- and school-level constructs to identify factors that may predict their performance. The results of PISA have been highly influential for policymakers across countries, leading them to adopt and adapt educational practices from high-performing systems to improve their own national educational systems (Breakspear, 2012). With mathematics designated as the major domain in the PISA 2022, the assessment focused on students’ mathematical literacy, and mathematics-related constructs were investigated to provide broader insights into students and schools (OECD, 2023). In particular, the newly introduced mathematical framework in the PISA 2022 explicitly emphasizes reasoning and real-life application, moving beyond the relatively narrow focus on traditional calculation tasks. In line with this shift, it is essential to explore ways of enhancing students’ mathematical achievement in real-life mathematical tasks.

Since the initial implementation of the PISA, numerous studies have explored the factors that predict students’ mathematics performance, and mathematics self-efficacy has consistently been identified as one of the most predictive and influential factors in explaining mathematics achievement (Gabriel et al., 2018; Huang et al., 2024; Kriegbaum et al., 2015; Lee & Stankov, 2018). Self-efficacy is defined as “people’s judgments of their capabilities to organize and execute courses of action required to attain designated types of performances” (Bandura, 1986, p. 391). In academic settings, mathematics self-efficacy refers to students’ confidence in their ability to successfully perform specific mathematical tasks, which influences their academic performance (Hackett & Betz, 1989; OECD, 2013; Schunk, 1991).

Despite the positive relationship between mathematics self-efficacy and mathematics achievement among students, many of the highest-performing countries in East Asia have exhibited low levels of mathematics self-efficacy. According to the PISA 2022 results, the six top-performing countries in mathematics—Singapore, Macao, Chinese Taipei, Hong Kong, Japan, and South Korea—are all located in East Asia. However, the countries’ average index scores for the two types of mathematics self-efficacy scales—the formal and applied mathematics self-efficacy and the mathematical reasoning and 21st century mathematics scales—were below the OECD (2024b) average, except for Singapore (for both scales) and Macao (for the formal and applied mathematics self-efficacy). Among these, Japan and South Korea recorded the lowest self-efficacy scores, which is consistent with findings from previous PISA cycles (Chiu, 2017; Lee, 2009; OECD, 2013). This result may appear paradoxical to policymakers who have sought to benchmark educational practices in these countries, given the well-documented positive link between self-efficacy and academic achievement. Prior studies have suggested that this phenomenon may be attributed to cultural influences such as Confucianism and collectivistic values, which, combined with a strong emphasis on meritocracy and high-stakes national exams, contribute to negative noncognitive outcomes among students regardless of their actual performance (Elliott et al., 2019; Kim, 2009; Stankov, 2010). From this perspective, the pursuit of high academic achievement does not necessarily lead to positive noncognitive outcomes. Specifically, aiming for exceptional mathematical performance should be accompanied by efforts to foster noncognitive factors, such as self-efficacy, to achieve more sustainable educational success. However, existing secondary analyses of the recent PISA 2022 dataset have mainly focused on identifying predictors of mathematics literacy and cognitive achievement (e.g., Huang et al., 2024; Niu et al., 2025; Pongsophon, 2025), whereas research examining predictors of noncognitive outcomes remains limited.

Research Purpose

This study aimed to explore the potential predictive factors of mathematics self-efficacy, with a particular focus on the newly introduced mathematics self-efficacy scale in the PISA 2022. Compared with the measures used in previous PISA cycles, students were asked about their confidence in handling 21st-century mathematics tasks. These tasks were not limited to simple arithmetic questions and operations, as has traditionally been the case; rather, they assessed students' confidence in critical thinking, problem-solving, and the practical application of mathematics in interdisciplinary contexts (OECD, 2024c). To the best of our knowledge, no previous studies have investigated the predictive factors associated with this new scale. Accordingly, this study utilized a wide variety of general and domain-specific constructs from the PISA survey to reconfirm previous findings on mathematics self-efficacy and identify new potential predictors, thereby contributing to the development of new hypotheses for future research. This study employed a regularization method, a commonly used machine learning approach in exploratory research, to select statistically significant variables from a large pool of candidates. However, previous studies that have applied regularization methods to large-scale assessment data have not adequately accounted for the nested structure of such datasets. For instance, the PISA uses a two-stage stratified sampling design to extract school and student samples and collects school- and student-level constructs to capture multiple layers of education settings. If the multilevel structure is ignored, the resulting estimates may be biased. Therefore, the present study adopted the generalized linear mixed model Lasso (glmmLasso), a machine learning method that enables Lasso regularization within a multilevel modeling framework (Yoo, 2024).

Furthermore, this study compared the findings from two high-performing East Asian countries, South Korea and Singapore, to examine the differences in the selected predictive constructs. Notably, Singapore was the only high-performing East Asian country in the PISA 2022 where students reported mathematics self-efficacy levels above the OECD average. In contrast, South Korea showed below-OECD-average mathematics self-efficacy levels regardless of mathematics performance level, with the largest gap in self-efficacy between low- and high-performing students among all participating countries (OECD, 2024b). Indeed, low-performing students in South Korea reported the lowest levels of mathematics self-efficacy across countries (OECD, 2024b). Examining two high-achieving countries with different mathematics self-efficacy levels within the same analytic framework allowed us to identify both common and country-specific predictors. Common predictive factors may offer implications applicable across diverse national contexts, whereas country-specific factors highlight how educational culture influences mathematics self-efficacy in distinct ways. Two research questions guided this study:

1. What are the significant factors that predict mathematics self-efficacy, as identified through the glmmLasso approach?
2. How do the types of selected predictive factors differ between South Korea and Singapore?

Theoretical Background

Mathematics Self-Efficacy

Mathematics self-efficacy refers to “a situational or problem-specific assessment of an individual’s confidence in her or his ability to successfully perform or accomplish a particular task or problem” (Hackett & Betz, 1989, p. 262). Self-efficacy plays a critical role in academic motivation and learning by shaping students’ decisions to engage in certain tasks, the effort they invest, and their persistence when faced with challenges (Bandura, 1997; Pajares, 1996; Schunk, 1991). Accordingly, students with high mathematics self-efficacy are more likely to undertake math-related tasks, exert greater effort, and persevere when solving difficult problems. Prior research has shown that mathematics self-efficacy is significantly associated not only with mathematics achievement (e.g., Lee et al., 2014; Pajares & Graham, 1999; Pajares & Miller, 1994) but also with noncognitive factors such as math interest, math anxiety, and mathematics-related career choices (e.g., Akin & Kurbanoglu, 2011; Betz & Hackett, 1983; Du et al., 2021; Lent et al., 1991, 1994). As both a key predictor and an outcome influenced by cognitive and noncognitive variables in academic settings, mathematics self-efficacy remains a central focus of educational research (Street et al., 2024; Zakariya, 2022).

Given that self-efficacy is not only domain-specific but also task-specific (Bandura, 1997; Pajares, 1996), self-efficacy can be distinguished from other self-belief constructs, particularly self-concept. Academic self-concept refers to individuals’ self-perceptions of competence formed within academic domains and is subject-specific, as in the case of mathematics self-concept (Bong & Skaalvik, 2003; Pajares & Miller, 1994; Shavelson et al., 1976). Both constructs are typically measured using self-report questionnaires. Self-concept is often assessed with items referring to the target subject, such as “I get good grades in mathematics,” “I have always believed that mathematics is one of my best subjects” (OECD, 2013), or “Mathematics is easy for me” (OECD, 2024c). In contrast, self-efficacy items ask about confidence level in performing

specific mathematical tasks, using prompts such as “How confident are you that you can successfully perform these tasks?” (Bong, 2006; see also Bong & Skaalvik, 2003; Hackett & Betz, 1989; Pajares & Miller, 1994). As mathematics self-efficacy is a context-specific assessment of student competence in a given mathematical task, it aligns with Bandura’s (1986) emphasis on establishing levels of specificity and correspondence to the criterial tasks (see also Bong, 2006; Pajares, 1996). For example, assessing mathematics self-efficacy using omnibus-type items that reflect students’ general confidence in mathematics may result in an obscure or redundant measure of the construct (Bandura, 1997; Pajares, 1996). In the PISA 2022, mathematics self-efficacy was measured using task-specific items targeting two distinct domains: traditional calculation with daily application and reasoning with 21st-century mathematics tasks. In line with the purpose of the present study, the analysis focused on the latter—reasoning and 21st-century mathematical tasks.

Factors Related to Mathematics Self-Efficacy

Previous research has examined the relationship between mathematics self-efficacy and various other factors. At the student level, gender has been consistently associated with mathematics self-efficacy, with male students tending to report higher levels of self-efficacy than female students (Lloyd et al., 2005; Recber et al., 2018; Usher et al., 2019; Wu, 2016). Higher economic, social, and cultural status scores have been positively associated with mathematics self-efficacy (Cuder et al., 2024; Wu, 2016). Additionally, parental academic support, expectations, and messages have been shown to play a significant role in shaping adolescents’ mathematics self-efficacy (Lee et al., 2022).

Students’ beliefs and emotional states have also been associated with mathematics self-efficacy. A proactive personality and high mathematics motivation are positively associated with self-efficacy (Lin et al., 2014; Özcan & Eren Gümüş, 2019). Feeling calm and emotionally secure during mathematics tasks is linked to a more positive perception of student competence (Usher & Pajares, 2009). In contrast, students who experience stress, fatigue, or emotional instability tend to report lower levels of mathematics self-efficacy (Usher & Pajares, 2009; Zakariya, 2022), and higher levels of mathematics anxiety have been shown to diminish self-efficacy (Cuder et al., 2024). Learning opportunities and instructional practices also play a role: increased exposure to both pure and applied mathematical tasks has been positively linked to students’ mathematics self-efficacy (Wang et al., 2024; Xiang et al., 2015). Among the various sources of self-efficacy, mastery experience is a highly important factor (Bandura, 1997; Zientek et al., 2019). Students interpret successful performances, particularly in challenging tasks, as evidence of their competence, whereas repeated failures tend to undermine their confidence (Usher & Pajares, 2009). Teacher support has also been associated with positive student outcomes in some studies (Wang et al., 2024); however, other findings suggest that social persuasion, such as teacher feedback and peer encouragement, may diminish rather than enhance mathematics self-efficacy (Lopez et al., 1997; Usher & Pajares, 2009; Yurt, 2014).

School-level factors have also been linked to students’ mathematics self-efficacy. Variables such as teacher–student ratio, class size, teacher motivation, and teacher morale have been reported to be associated with mathematics self-efficacy (Usta, 2016). Moreover, both teacher training and teacher-centered classroom climates have been identified as factors related to higher mathematics self-efficacy (Peters, 2013; Siegle & McCoach, 2007).

Methods

Research Design

This study employed an exploratory research design to identify potential predictors of students’ mathematics self-efficacy in the large-scale PISA 2022 dataset using the machine learning-based method glmLasso. Comparing South Korea and Singapore, two high-achieving countries that differ in levels of mathematics self-efficacy, we examined which predictors were common across the two countries and which were country-specific.

Data

The data for this study were obtained from the South Korean and Singapore subsets of the PISA 2022. The PISA employs a two-stage stratified sampling design: schools with 15-year-old students were sampled as primary units, and students were sampled within these selected schools, resulting in a nested data structure (OECD, 2024c). Some samples were excluded because of missing values on the mathematics self-efficacy scale or because they exhibited a relatively high missing rate of over 20% across the total initially adopted variables. After the data tidying process, the final analytical sample comprised 6,300 students from 184 schools in Korea and 6,512 students from 164 schools in Singapore.

Measures

In addition to assessing the mathematics literacy of students near the completion of compulsory schooling, the PISA administered a student background questionnaire and a school questionnaire in all participating countries, resulting in two mandatory core context questionnaires. Through these background questionnaires, PISA collected various types of information from students and schools, which enabled researchers to explore both student- and school-level constructs that lead to effective educational outcomes. In the PISA 2022, these variables are categorized into a two-dimensional framework: one dimension classifies variables into domain-specific constructs and general constructs, and the other classifies variables into five categories based on the key policy focus areas (student background; student beliefs, attitudes, feelings, and behaviors; teaching practices and learning opportunities; school practices, policies, and infrastructure; and governance, system-level policies, and practices) (OECD, 2023; 2024c). Within this framework, constructs from both student and school questionnaires were utilized as candidate variables to predict mathematics self-efficacy, with a focus on reasoning and 21st-century mathematical tasks.

The initial set of variables was derived from the PISA 2022, with variables constructed either through simple arithmetic calculations or as scaled indices based on item response theory (IRT). However, to fully align with the modules covered in the PISA 2022 context questionnaire and framework, some variables were manually aggregated into sum scores by the researchers. For example, a construct that measured schools' provision of information about future educational/work/career path (SC171), originally composed of four binary items, was aggregated into a single sum score. Variables unrelated to the modules were excluded, and only a complex composite index was retained to measure students' economic, social, and cultural status (ESCS).

As an outcome variable, mathematics self-efficacy, focusing on reasoning and 21st-century mathematics tasks, was assessed through the common prompt, "How confident do you feel about having to do the following mathematics tasks?" using 10 items rated on a 4-point scale, where 1 indicated "not at all confident" and 4 indicated "very confident" (OECD, 2024c). Compared to the previously administered mathematics self-efficacy scale in the PISA, this scale targets mathematical tasks related to "interpretation and analysis of mathematical data; real-world application; statistical reasoning; mathematical modeling; computational and technology literacy; and geometry and measurement" (OECD, 2024b). Example items include "Interpreting mathematical solutions in the context of a real-life challenge" and "Working with computer mathematics systems (e.g., spreadsheets, programming software, graphing calculators)" (OECD, 2024c). To accommodate this broad item pool, the PISA implemented a within-construct matrix sampling design in which each student responded to only five randomly selected items. This design allowed for an in-depth assessment of the construct across a wide range of tasks while minimizing the respondent burden. Therefore, this study used an IRT-based scaled mathematics self-efficacy score, estimated using a weighted likelihood estimate with senate weights and further standardized to have a mean of 0 and standard deviation (SD) of 1 across OECD countries.

Data Preprocessing

Variables with more than 20% missing values at either the student or school level were excluded from the initially selected constructs. The remaining missing values were imputed using the k -nearest neighbor (k -NN) method, which has been widely applied in predictive research (Yoo & Rho, 2020, 2022). In addition, zero-variance variables and one variable from each pair with a bivariate correlation coefficient exceeding .90 were excluded. To further address multicollinearity, variables with variance inflation factor values greater than 10 were sequentially removed. The final set of candidate variables for predicting mathematics self-efficacy comprised 52 student-level and 56 school-level constructs. Detailed descriptions of the selected variables, including those constructed as manually computed composite scores, are provided in the [Appendix](#).

Analysis

This study applied glmLasso, a machine learning method based on regularization techniques, to identify the potential predictors of mathematics self-efficacy from a large set of candidate variables, followed by multilevel linear modeling to test their statistical significance. Regularization typically incorporates a penalty term into the regression equation, thereby offering the advantages of parsimony and overfitting prevention, making it particularly useful in social science research (Yoo & Rho, 2020). Among the regularization methods, Lasso is one of the most widely recognized techniques because it shrinks certain coefficients to zero, thereby simultaneously performing variable selection and coefficient estimation. However, this approach is limited in that it does not account for hierarchical data structures. To address this limitation, this study employed glmLasso, which accommodates the multilevel structure of the data (Groll & Tutz, 2014).

Understanding glmLasso requires prior knowledge of the generalized linear mixed model, as shown in Equation (1):

$$g(\mu) = X\beta + Zb, b \sim \mathcal{N}(0, Q) \quad (1)$$

where the parameter μ represents the conditional mean of the outcome variable y . The function $g(\mu)$ denotes the link function, for which the identity function was used in this study. Matrix X represents the fixed-effects design matrix across all observations, whereas Z is a block diagonal matrix comprising the random-effects design matrices of each cluster. The coefficient vector β represents the fixed effects, while the random-effects coefficient vector b is assumed to follow a multivariate normal distribution with a mean of zero and a block-diagonal covariance matrix Q .

GlmmLasso applies generalized linear mixed models with L1-penalization (Tibshirani, 1996) to perform fixed-effects selection and parameter estimation in the presence of multilevel structures, as shown in Equation (2):

$$\hat{\delta} = \underset{\delta}{\operatorname{argmax}} l^{\text{pen}}(\delta, \hat{\gamma}) = \underset{\delta}{\operatorname{argmax}} \left[l^{\text{app}}(\delta, \hat{\gamma}) - \lambda \sum_{i=1}^p |\beta_i| \right] \quad (2)$$

where δ is the combined vector of fixed and random effects (β and b), $\hat{\gamma}$ includes the dispersion and random effects variance parameters, and λ is the L1-penalty parameter controlling the degree of shrinkage (Groll & Tutz, 2014). The log-likelihood was obtained via the Laplace approximation, which is a method for approximating complex integrals (Breslow & Clayton, 1993), denoted as l^{app} . The penalized log-likelihood is given by l^{pen} , which is derived by adding a penalty term to the Laplace approximation l^{app} . The estimation procedure followed a gradient ascent algorithm (Goeman, 2010) using a second-order Taylor approximation of the penalized likelihood. At each iteration, the algorithm updated the parameters along the penalized gradient direction with an adaptive step size, ensuring convergence within the constrained parameter space (Groll & Tutz, 2014).

The procedure for this analysis is as follows. First, the glmmLasso function was applied, with mathematics self-efficacy as the outcome variable and 108 variables as predictors. The raw data were randomly split at the school level into training and test datasets in an 8:2 ratio, and this process was repeated 100 times, resulting in 100 data pairs. All continuous variables in the training data were standardized to $SD = 1$, and the test data were scaled using the mean and SD computed from the training data to prevent data leakage. The training datasets were used for variable selection across the 100 resampled datasets to enhance the robustness of the results. Unlike the traditional Lasso implemented in the glmnet package in R, glmmLasso does not automatically compute the optimal lambda through cross-validation (Bollmann et al., 2023). Therefore, to determine the lambda grid, we identified the lambda value at which no variables were selected and searched the subsequent five grid points for lambda values at which at least one variable was selected (Bollmann et al., 2023). Accordingly, the lambda range was set from 2100 to 0 and divided into 200 grid points, decreasing by approximately 10 each time. For each training dataset, the lambda that minimized the Bayesian information criterion (BIC) was selected as the optimal lambda, and variables with coefficients that did not shrink to zero at that lambda were retained. BIC is a widely used model selection criterion that balances model fit and complexity by penalizing the number of parameters and the sample size (Schwarz, 1978). A smaller BIC value indicates a better fit. The Akaike information criterion (AIC) was used as a supplementary consistency check with the BIC results. Variables selected in 50% or more of the iterations were considered stable predictors (e.g., Yoo & Rho, 2020), incorporating the concepts of relevance counts (Shevade & Keerthi, 2003) and stability selection (Meinshausen & Bühlmann, 2010). The test datasets were used to calculate the root mean square error (RMSE) and its SD to assess predictive performance and estimate stability across iterations.

Next, a multilevel linear model was fitted to the full dataset to examine the statistical significance of the variables selected using glmmLasso. Rather than relying on the internal re-estimation procedure embedded in the glmmLasso function, a multilevel linear model with a random intercept and fixed slopes was employed using PISA sampling weights to ensure accurate coefficient estimates and alignment with standard practices in international large-scale assessment studies, with estimation performed using restricted maximum likelihood. All analyses were conducted in R 4.5.0 using the glmmLasso and lme4 packages.

Results

glmmLasso Results

As a preliminary analysis, the intraclass correlation coefficient (ICC) was calculated to examine the multilevel structure of mathematics self-efficacy. The ICC was calculated using a null model without independent variables as the ratio of between-group variance to total variance (i.e., between-group variance + within-group variance) (Hox et al., 2017). The ICC for mathematics self-efficacy was 0.089 in South Korea and 0.065 in Singapore, indicating that approximately 8.9% and 6.5% of the total variance in each country, respectively, was attributable to differences between schools.

Table 1 presents descriptive statistics summarizing the glmLasso results across 100 iterations, using different training and test datasets from South Korea and Singapore. Among the 108 candidate variables for predicting mathematics self-efficacy, an average of 23.28 and 22.38 variables showed non-zero coefficients per iteration in the models for South Korea and Singapore, respectively. The mean optimal lambda values across iterations were 139.19 for South Korea and 134.97 for Singapore. The corresponding mean AIC values were 10693.84 and 11852.60, and the mean BIC values were 10999.37 and 12214.49, respectively. The average RMSE values for South Korea and Singapore were 0.7285 (SD = 0.0181) and 0.7754 (SD = 0.0209), respectively.

Table 2 summarizes the selection frequency of variables across 100 iterations. Among the 108 candidate variables, only 13 in South Korea and 11 in Singapore were selected in all 100 iterations. Additionally, three variables were selected in more than 80 iterations and six in more than 50 iterations for both South Korea and Singapore; however, the specific variables differed between the two countries. Based on the glmLasso selection process, a total of 22 variables for South Korea and 20 variables for Singapore were selected in 50% or more of the iterations and retained as the final set of candidate variables.

Table 3 presents the final variable sets for South Korea and Singapore that were applied to the subsequent multilevel linear model. Constructs selected in both countries included economic, social, and cultural status (ESCS), gender (SEX), mathematics anxiety (ANXMAT), proactive mathematics study behavior (MATHPERS), creative thinking self-efficacy (CREATEFF), creativity and openness to intellect (CREATOP), perseverance (PERSEVAGR), stress resistance (STRESAGR), curiosity (CURIOAGR), highest expected education level (EXPECEDU), exposure to mathematical reasoning and 21st-century mathematics tasks (EXPO21ST), subjective familiarity with mathematics concepts (FAMCON), cognitive activation in mathematics: encourage mathematical thinking (COGACMCO), and openness to art and reflection (OPENART).

Eight student-level variables were selected exclusively for South Korea: time spent on mathematics homework (ST296Q01JA), studying for school or homework before or after school (STUDYHMW), cooperation (COOPAGR), mathematics teacher support (TEACHSUP), exposure to formal and applied mathematics tasks (EXPOFA), disciplinary climate in mathematics (DISCLIM), creative activities at school (CREATAS), and creative activities outside of school (CREATOOS). For Singapore, four student-level variables were uniquely selected—assertiveness (ASSERAGR), exercising or practicing a sport before or after school (EXERPRAC), time spent on online activities (ST326_mean), and creative school and class environment (CREATSCH)—along with two school-level variables—teacher's ISCED level 8 doctoral qualification (PROPAT8) and information about future educational/work/career path (SC171_mean). Notably, no school-level variables were selected for South Korea.

Multilevel Linear Modeling Results

Table 4 presents the results of the multilevel linear model for the variables retained through glmLasso. Several factors exhibited consistent effects on mathematics self-efficacy in both South Korea and Singapore. At the student level, socioeconomic status (ESCS) in the student background domain positively predicted mathematics self-efficacy in both countries ($\beta = 0.054$ in Korea, $\beta = 0.059$ in Singapore, $p < .001$). Gender (SEX) also showed a significant effect, with male students reporting higher levels of self-efficacy than female students ($\beta = 0.185$ in Korea, $\beta = 0.248$ in Singapore, $p < .001$).

In the domain of student beliefs, attitudes, and behaviors, mathematics self-efficacy was negatively predicted by mathematics anxiety (ANXMAT) ($\beta = -0.186$ in Korea, $\beta = -0.170$ in Singapore, $p < .001$) and positively predicted by proactive mathematics study behavior (MATHPERS) ($\beta = 0.194$ in Korea, $\beta = 0.108$ in Singapore, $p < .001$). Creative thinking self-efficacy (CREATEFF) ($\beta = 0.111$ in Korea, $\beta = 0.100$ in Singapore, $p < .001$), creativity and openness to intellect (CREATOP) ($\beta = 0.092$ in Korea, $\beta = 0.072$ in Singapore, $p < .001$), and expected educational level (EXPECEDU) ($\beta = 0.030$, $p < .001$ in Korea; $\beta = 0.019$, $p < .01$ in Singapore) also positively predicted mathematics self-efficacy in both countries. Curiosity (CURIOAGR) ($\beta = 0.030$ in Korea, $\beta = 0.024$ in Singapore, $p < .05$), perseverance (PERSEVAGR) ($\beta = 0.057$ in Korea, $p < .001$; $\beta = 0.031$ in Singapore, $p < .01$), and stress resistance (STRESAGR) ($\beta = 0.040$ in Korea, $p < .01$;

Table 1. Descriptive Statistics of glmLasso Results Across 100 Iterations

Country	# of selected variables	Optimal lambda (λ)	AIC	BIC	RMSE (Std. Dev.)
South Korea	23.28	139.19	10693.84	10999.37	0.7285 [0.0181]
Singapore	22.38	134.97	11852.60	12214.49	0.7754 [0.0209]

Note. AIC: Akaike information criterion; BIC: Bayesian information criterion; RMSE: root mean square error.

Table 2. Relevance Counts (*n*) of Variables Across 100 glmmLasso Iterations

Country	<i>n</i> = 100	<i>n</i> ≥ 80	<i>n</i> ≥ 50	<i>n</i> ≥ 1	<i>n</i> = 0
South Korea	13	3	6	39	47
Singapore	11	3	6	35	53

$\beta = 0.067$ in Singapore, $p < .001$) were positively associated with mathematics self-efficacy. In contrast, openness to art and reflection (OPENART) had a negative association ($\beta = -0.052$ in Korea, $p < .001$; $\beta = -0.033$ in Singapore, $p < .01$).

In the domain of teaching practices and learning opportunities, exposure to mathematical reasoning and 21st-century tasks (EXPO21ST) ($\beta = 0.104$ in Korea, $\beta = 0.138$ in Singapore, $p < .001$), subjective familiarity with mathematics concepts (FAMCON) ($\beta = 0.131$ in Korea, $\beta = 0.144$ in Singapore, $p < .001$), and cognitive activation in mathematics (COGACMCO) ($\beta = 0.053$ in Korea, $\beta = 0.055$ in Singapore, $p < .001$) were all positively associated with mathematics self-efficacy in both countries.

Some factors exhibited country-specific effects. In South Korea, within the domain of student beliefs, attitudes, and behaviors, time spent on mathematics homework (ST296Q01JA) ($\beta = 0.048$, $p < .001$) and studying for school or homework before or after school (STUDYHMW) ($\beta = 0.018$, $p < .001$) positively predicted mathematics self-efficacy. In contrast, cooperation (COOPAGR) ($\beta = -0.070$, $p < .001$) and mathematics teacher support (TEACHSUP) ($\beta = -0.038$, $p < .001$) had negative effects. In Singapore, time spent on online activities (ST326_mean) ($\beta = 0.018$, $p < .05$) had a positive effect on mathematics self-efficacy, while assertiveness (ASSERAGR) and exercising or practicing a sport before or after school (EXERPRAC) showed no significant effects.

In South Korea, regarding teaching practices and learning opportunities variables, exposure to formal and applied mathematics tasks (EXPOFA) ($\beta = 0.039$, $p < .001$) and creative activities at school (CREATAS) ($\beta = 0.044$, $p < .01$) positively predicted mathematics self-efficacy, whereas disciplinary climate in mathematics (DISCLIM) negatively

Table 3. Final Sets of glmmLasso-Selected Variables for South Korea and Singapore

Category		South Korea		Singapore	
Level	Policy area	Variable	Relevance counts	Variable	Relevance counts
Student-level	Student Background	ESCS	100	ESCS	100
		SEX	80	SEX	100
	Student Beliefs, Attitudes, Feelings, and Behaviors	ANXMAT	100	ANXMAT	100
		MATHPERS	100	MATHPERS	100
		CREATEFF	100	CREATEFF	100
		CREATOP	100	CREATOP	100
		PERSEVAGR	100	STRESAGR	100
		EXPECEDU	100	PERSEVAGR	99
		CURIOAGR	92	CURIOAGR	98
		STRESAGR	90	EXPECEDU	90
		OPENART	59	OPENART	59
		ST296Q01JA	100	ASSERAGR	78
		STUDYHMW	100	EXERPRAC	60
		COOPAGR	76	ST326_mean	57
		TEACHSUP	60		
	Teaching Practices and Learning Opportunities	EXPO21ST	100	EXPO21ST	100
		FAMCON	100	FAMCON	100
		COGACMCO	100	COGACMCO	100
		EXPOFA	100	CREATSCH	100
		CREATOOS	61		
		DISCLIM	60		
		CREATAS	52		
School-level	School Practices, Policies, and Infrastructure			PROPAT8	71
				SC171_mean	53

Table 4. Multilevel Linear Modeling Results for South Korea and Singapore

Category		South Korea		Singapore	
Level	Policy area	Variable	Estimate (Std. Error)	Variable	Estimate (Std. Error)
Student-level	Student Background	ESCS ^{***}	0.054(0.015)	ESCS ^{***}	0.059 (0.012)
		SEX (Male) ^{***}	0.185 (0.023)	SEX (Male) ^{***}	0.248 (0.021)
	Student Beliefs, Attitudes, Feelings, and Behaviors	ANXMAT ^{***}	-0.186 (0.009)	ANXMAT ^{***}	-0.170 (0.009)
		MATHPERS ^{***}	0.194 (0.012)	MATHPERS ^{***}	0.108 (0.012)
		CREATEFF ^{***}	0.111 (0.013)	CREATEFF ^{***}	0.100 (0.012)
		CREATOP ^{***}	0.092 (0.014)	CREATOP ^{***}	0.072 (0.013)
		EXPECEDU ^{***}	0.030 (0.006)	EXPECEDU ^{**}	0.019 (0.007)
		CURIOAGR [*]	0.030 (0.012)	CURIOAGR [*]	0.024 (0.011)
		PERSEVAGR ^{***}	0.057 (0.012)	PERSEVAGR ^{**}	0.031 (0.011)
		STRESAGR ^{**}	0.040 (0.013)	STRESAGR ^{***}	0.067 (0.011)
		OPENART ^{***}	-0.052 (0.013)	OPENART ^{**}	-0.033 (0.012)
		ST296Q01JA ^{***}	0.048 (0.009)	ASSERAGR	0.016 (0.010)
		STUDYHWM ^{***}	0.018 (0.004)	EXERPRAC	-0.001 (0.003)
		COOPAGR ^{***}	-0.070 (0.013)	ST326_mean [*]	0.018 (0.007)
		TEACHSUP ^{***}	-0.038 (0.011)		
	Teaching Practices and Learning Opportunities	EXPO21ST ^{***}	0.104 (0.012)	EXPO21ST ^{***}	0.138 (0.011)
		FAMCON ^{***}	0.131 (0.007)	FAMCON ^{***}	0.144 (0.007)
		COGACMCO ^{***}	0.053 (0.011)	COGACMCO ^{***}	0.055 (0.011)
		EXPOFA ^{***}	0.039 (0.008)	CREATSCH ^{***}	0.046 (0.011)
		CREATOOS	0.004 (0.014)		
		DISCLIM ^{***}	-0.043 (0.012)		
		CREATAS ^{**}	0.044 (0.015)		
School-level	School Practices, Policies, and Infrastructure			PROPAT8 ^{**}	1.936 (0.656)
				SC171_mean ^{**}	0.038 (0.011)

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

predicted it ($\beta = -0.043, p < .001$). Creative activities outside of school (CREATOOS) was not significantly associated with mathematics self-efficacy. In Singapore, creative school and class environment (CREATSCH) ($\beta = 0.046, p < .001$) was positively associated with mathematics self-efficacy.

At the school level, in Singapore, both the proportion of teachers with doctoral qualifications (PROPAT8) ($\beta = 1.936, p < .01$) and provision of career-related information (SC171_mean) ($\beta = 0.038, p < .01$) positively predicted students' mathematics self-efficacy. No school-level variables were selected for South Korea.

Discussion

This study examined the predictors of mathematics self-efficacy using 108 candidate variables from the PISA 2022 dataset. Variable selection was conducted using glmLasso, and the statistical significance of the selected variables was tested using a multilevel linear model. The analysis identified 21 student-level predictors for South Korea and 16 student-level and two school-level predictors for Singapore.

Common Predictors

At the student level, higher ESCS and being male (SEX) were associated with higher mathematics self-efficacy, consistent with prior research (Cadinu et al., 2003; Good et al., 2008; Lee et al., 2022; Rosenthal et al., 2007; Steele, 1997; Wu, 2016). These findings suggest the need for efforts to enhance mathematics self-efficacy among female students and those from lower socioeconomic backgrounds, with the aim of reducing disparities associated with social background.

Within the domain of students' beliefs, attitudes, and behaviors, several variables were associated with mathematics self-efficacy. Higher levels of mathematics anxiety (ANXMAT) were associated with lower levels of self-efficacy, consistent with evidence that anxious feelings may generate negative emotional arousal that contributes to lower academic confidence (Bandura, 1977). This result is consistent with previous studies indicating that students' high anxiety around mathematics

(Du et al., 2021; Griggs et al., 2013) and negative emotions toward mathematics (Usher & Pajares, 2009; Zakariya, 2022) are significantly associated with low mathematics self-efficacy. Conversely, proactive mathematics study behavior (MATHPERS) was positively associated with mathematics self-efficacy, consistent with prior findings that students with proactive personalities tend to exhibit higher self-efficacy (Lin et al., 2014). Among social and emotional characteristics, perseverance (PERSEVAGR) and stress resistance (STRESAGR) were also positively associated with mathematics self-efficacy, in line with research linking positive psychological resources such as grit to stronger mathematics self-efficacy (Usher et al., 2019; Zakariya, 2022).

Curiosity (CURIOAGR), creative thinking self-efficacy (CREATEFF), and creativity and openness to intellect (CREATOP) were all positively associated with mathematics self-efficacy. Although these constructs are conceptually distinct—curiosity refers to a general socioemotional characteristic, creative thinking self-efficacy denotes students' belief in their ability to perform creative work in challenging situations, and openness to intellect constitutes a personality trait reflecting an inclination toward creativity—their conceptual relatedness suggests that mathematics self-efficacy, in the context of mathematical reasoning and 21st-century tasks, may be closely linked to curiosity and creative orientation (OECD, 2024a). This pattern is particularly notable given that East Asian schooling systems are frequently organized around examination-based selection for university admission, a high-stakes environment where students' motivation often stems less from a genuine desire for knowledge and more from university acceptance and future employment. This absence of intellectual curiosity is closely linked to a lack of creativity in education (Kim, 2009), which may contribute to students' low self-efficacy. Future research should explore the interplay among curiosity, creativity, and self-efficacy in mathematics in more detail. As creativity grows in importance in the 21st century (OECD, 2024a), mathematics instruction should be designed to encourage students' curiosity and foster creativity within the classroom—not only at macroscopic policy levels but also in everyday teaching and learning contexts.

By contrast, openness to art and reflection (OPENART) negatively predicted mathematics self-efficacy, whereas expected educational level (EXPECEDU) showed a positive association. As these variables emerged through this study's exploratory approach and have not been extensively examined in previous research, further studies are required to clarify their relationships with mathematics self-efficacy.

In the domain of teaching practices and learning opportunities, exposure to mathematical reasoning and 21st-century tasks (EXPO21ST), subjective familiarity with mathematics concepts (FAMCON), and cognitive activation in mathematics (COGACMCO) were associated with higher mathematics self-efficacy in both countries. These findings align with previous research demonstrating that greater learning opportunities enhance mathematics self-efficacy (Wang et al., 2024) and suggest that increasing students' exposure to tasks involving critical thinking, problem-solving, and practical mathematics application may enhance their mathematics self-efficacy in addition to providing opportunities for cognitive activation (OECD, 2024b). Based on this study's findings, future research should conduct more detailed analyses examining the mediating or moderating roles of associated factors, such as the psychological traits identified in this study (e.g., MATHPERS and CURIOAGR), in the relationship between instructional exposure and mathematics self-efficacy.

Country-Specific Predictors

Next, we discuss the differences between the two countries and country-specific predictors of mathematics self-efficacy. South Korea operates under a nationally standardized curriculum in which admission to higher education is based on performance on norm-referenced assessments (Chun et al., 2019). University entrance exams emphasize efficient problem-solving, requiring students to complete many items under strict time constraints (Byun et al., 2012). Several country-specific predictors can be understood within this context. Time spent on mathematics homework (ST296Q01JA), studying for school or homework before or after school (STUDYHMW), and exposure to formal and applied mathematics tasks (EXPOFA) were positively associated with mathematics self-efficacy. In an educational context that emphasizes repetition and mastery, increased engagement with and exposure to mathematical tasks likely strengthens self-efficacy through accumulated mastery experiences. Prior research has identified mastery experience as a highly important factor associated with self-efficacy, as identified by Bandura (1997; see also Zientek et al., 2019).

However, cooperation (COOPAGR) negatively predicted mathematics self-efficacy in South Korea. Prior studies have suggested that high-stakes national exams contribute to negative noncognitive outcomes regardless of students' academic achievement (Elliott et al., 2019; Kim, 2009; Stankov, 2010). In South Korea's norm-referenced assessment environment, increasing individual study time is often perceived as a more efficient learning strategy than collaboration. Consequently, lower collaboration levels may be associated with higher mathematics self-efficacy, which aligns with the tendency for high-achieving students to prioritize individual study for efficiency. Similarly, mathematics teacher support (TEACHSUP) was negatively associated with mathematics self-efficacy, a finding that may reflect the tendency for teacher support in South Korea to focus on low-performing students, who are also more likely to exhibit low mathematics self-efficacy. Moreover,

consistent with Bandura's (1977) statement that the effect of social persuasion depends on how feedback is perceived, students who receive more teacher support may interpret it as a signal of lower ability, which may, in turn, suppress mathematics self-efficacy. Finally, a more disciplinary climate in mathematics (DISCLIM) was also negatively associated with mathematics self-efficacy. Collectively, these results suggest that a more supportive, student-centered environment has the potential to enhance students' mathematics self-efficacy.

In Singapore, time spent on online activities was significantly associated with mathematics self-efficacy, which may reflect the country's well-established digital learning infrastructure (Kong et al., 2014; Powell & Barbour, 2018). This finding suggests that strengthening online platforms may enhance access to learning resources for students in need of support. Such accessibility could facilitate the development of mathematics skills and, consequently, mathematics self-efficacy.

Regarding creativity, engagement in creative activities at school (CREATAS) in South Korea and a creative school and classroom environment (CREATSCH) in Singapore were significantly positively associated with mathematics self-efficacy. Although the two variables differ slightly in name and definition, they reinforce the broader findings that expanding creativity-supporting classroom activities and learning environments may be a practical way to foster students' mathematics self-efficacy.

At the school level, the proportion of teachers with doctoral qualifications (PROPAT8) was positively associated with students' mathematics self-efficacy in Singapore. This finding appears to differ slightly from previous research suggesting that teacher qualifications are not significantly related to self-efficacy (Saenz et al., 2023). In addition, the provision of information about future education, work, and career pathways (SC171_mean) was positively associated with students' mathematics self-efficacy. This result aligns with evidence that career guidance and career awareness can support STEM self-efficacy (Han et al., 2021), suggesting that career education may play an important role in fostering mathematics self-efficacy. These two findings may reflect features of Singapore's education system. The country provides structured support for teachers' professional development, including pathways such as master's and doctoral programs (Bautista et al., 2015). In addition, Singapore has a well-established, school-based career guidance system (Wong & Liem, 2025). Schools with highly qualified teachers and effective career guidance may create broader institutional conditions conducive to positive student outcomes, including higher mathematics self-efficacy. However, because the present findings were derived through an exploratory procedure, they should not be interpreted as evidence of a causal relationship with mathematics self-efficacy. In South Korea, no school-level variables were included in the multilevel linear model as none met the 50% cutoff criterion for selection in glmLasso. School-level variables, such as the teacher–student ratio and class size, were not significantly associated with mathematics self-efficacy in either country.

Limitations

This study had several limitations. First, although it was conducted using student- and school-level factors based on the PISA 2022 analytical framework, it remains inherently exploratory because of the application of a data-driven machine learning approach (glmLasso). Therefore, the results should be interpreted with caution. Second, although common predictors were identified across the two countries, suggesting a certain degree of generalizability, these findings were derived from an East Asian context and should be interpreted carefully when applied to other cultural settings. Third, variables that were found to be significant at the student level may exhibit contextual effects when aggregated at the school level, resulting in constructs such as classroom climate or average ESCS. Future studies should explore the potential contextual effects.

Conclusion and Implications

This study examined factors associated with students' mathematics self-efficacy in South Korea and Singapore using PISA 2022 data. Across both countries, personal characteristics such as perseverance and stress resistance were positively associated with mathematics self-efficacy. Learning opportunities, such as cognitively activating instruction and exposure to 21st-century mathematics tasks, were positively associated with higher self-efficacy, as were curiosity, creativity, and a creative educational environment. Regarding country-specific variables, in South Korea, individual learning factors—such as time spent on mathematics tasks and after-school study—were positively associated with mathematics self-efficacy, whereas collaboration and teacher support were negatively associated. In contrast, in Singapore, students' use of online learning was positively associated with mathematics self-efficacy, and at the school level, both the proportion of teachers with doctoral qualifications and the provision of future education and career information were positively associated.

This study is significant in that it examined mathematics self-efficacy related to reasoning and 21st-century mathematical tasks using the newly released PISA 2022 data. By identifying common predictors in two high-achieving countries with different levels of mathematics self-efficacy, this study provides insights into improving mathematics self-efficacy. Additionally, country-specific predictors were explored, revealing that educational contexts are differently associated with mathematics self-efficacy across countries. Building on the exploratory variables identified in this study, future research should conduct in-depth investigations to enhance students' mathematics self-efficacy in the current educational landscape.

ORCID iDs

Myongsung Lee  <https://orcid.org/0000-0002-9323-3289>

Seongmin Park  <https://orcid.org/0009-0008-5941-1329>

Sehee Hong  <https://orcid.org/0000-0001-5468-8398>

Ethical Considerations

Ethics approval was not required for this study as it used publicly available secondary data from the PISA 2022 survey conducted by the OECD.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The data used in this study are publicly available from the OECD's Programme for International Student Assessment (PISA) 2022 database: <https://www.oecd.org/pisa/data/2022database>.

References

- Akin, A., & Kurbanoglu, I. N. (2011). The relationships between math anxiety, math attitudes, and self-efficacy: A structural equation model. *Studia Psychologica*, 53(3), 263–273.
- Bandura, A. (1977). Self-efficacy: toward a unifying theory of behavioral change. *Psychological Review*, 84(2), 191–215. <https://doi.org/10.1037//0033-295x.84.2.191>
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice-Hall.
- Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman.
- Bautista, A., Wong, J., & Gopinathan, S. (2015). Teacher professional development in Singapore: Depicting the landscape. *Psychology, Society, & Education*, 7(3), 311–326. <https://doi.org/10.25115/psye.v7i3.523>
- Betz, N. E., & Hackett, G. (1983). The relationship of mathematics self-efficacy expectations to the selection of science-based college majors. *Journal of Vocational Behavior*, 23(3), 329–345. [https://doi.org/10.1016/0001-8791\(83\)90046-5](https://doi.org/10.1016/0001-8791(83)90046-5)
- Bollmann, S., Groll, A., & Havranek, M. M. (2023). Accounting for clustering in automated variable selection using hospital data: a comparison of different LASSO approaches. *BMC Medical Research Methodology*, 23(1), 280. <https://doi.org/10.1186/s12874-023-02081-6>
- Bong, M. (2006). Asking the right question: How confident are you that you could successfully perform these tasks. In F. Pajares & T. Urdan (Eds.), *Self-efficacy beliefs of adolescents* (pp. 287–305). Information Age Publishing.
- Bong, M., & Skaalvik, E. M. (2003). Academic self-concept and self-efficacy: How different are they really? *Educational Psychology Review*, 15, 1–40. <https://doi.org/10.1023/a:1021302408382>
- Breakspear, S. (2012). *The policy impact of PISA: An exploration of the normative effects of international benchmarking in school system performance (OECD Education Working Papers, No. 71)*. OECD Publishing. <https://doi.org/10.1787/5k9fdqffr28-en>
- Breslow, N. E., & Clayton, D. G. (1993). Approximate inference in generalized linear mixed models. *Journal of the American Statistical Association*, 88(421), 9–25. <https://doi.org/10.1080/01621459.1993.10594284>
- Byun, S. Y., Schofer, E., & Kim, K. K. (2012). Revisiting the role of cultural capital in East Asian educational systems: The case of South Korea. *Sociology of Education*, 85(3), 219–239. <https://doi.org/10.1177/0038040712447180>

- Cadinu, M., Maass, A., Frigerio, S., Impagliazzo, L., & Latinotti, S. (2003). Stereotype threat: The effect of expectancy on performance. *European journal of social psychology*, 33(2), 267–285. <https://doi.org/10.1002/ejsp.145>
- Chiu, M. M. (2017). Self-concept, self-efficacy, and mathematics achievement: students in 65 regions including the US and Asia. In: *What matters? Research trends in international comparative studies in mathematics education* (pp. 267–288). Springer.
- Chun, H., Son, H., & Woo, J. Y. (2019). Lived experiences of high school students for the “Naesin” grading as a norm-referenced evaluation. *The Korean Journal of School Psychology*, 16(3), 401–431.
- Cuder, A., Pellizzoni, S., Di Marco, M., Blason, C., Doz, E., Giofrè, D., & Passolunghi, M. C. (2024). The impact of math anxiety and self-efficacy in middle school STEM choices: A 3-year longitudinal study. *British Journal of Educational Psychology*, 94(4), 1091–1108. <https://doi.org/10.1111/bjep.12707>
- Davis, P. J. (1988). Applied mathematics as social contract. *Mathematics Magazine*, 61(3), 139–147. <https://doi.org/10.1080/0025570x.1988.11977361>
- Du, C., Qin, K., Wang, Y., & Xin, T. (2021). Mathematics interest, anxiety, self-efficacy and achievement: Examining reciprocal relations. *Learning and Individual Differences*, 91, 102060. <https://doi.org/10.1016/j.lindif.2021.102060>
- Elliott, J., Stankov, L., Lee, J., & Beckmann, J. F. (2019). What did PISA and TIMSS ever do for us? The potential of large scale datasets for understanding and improving educational practice. *Comparative Education*, 55(1), 133–155. <https://doi.org/10.1080/03050068.2018.1545386>
- Gabriel, F., Signolet, J., & Westwell, M. (2018). A machine learning approach to investigating the effects of mathematics dispositions on mathematical literacy. *International Journal of Research & Method in Education*, 41(3), 306–327. <https://doi.org/10.1080/1743727x.2017.1301916>
- Goeman, J. J. (2010). L1 penalized estimation in the Cox proportional hazards model. *Biometrical Journal*, 52(1), 70–84. <https://doi.org/10.1002/bimj.200900028>
- Good, C., Aronson, J., & Harder, J. A. (2008). Problems in the pipeline: Stereotype threat and women's achievement in high-level math courses. *Journal of Applied Developmental Psychology*, 29(1), 17–28. <https://doi.org/10.1016/j.appdev.2007.10.004>
- Griggs, M. S., Rimm-Kaufman, S. E., Merritt, E. G., & Patton, C. L. (2013). The Responsive Classroom approach and fifth grade students' math and science anxiety and self-efficacy. *School Psychology Quarterly*, 28(4), 360–373. <https://doi.org/10.1037/spq0000026>
- Groll, A., & Tutz, G. (2014). Variable selection for generalized linear mixed models by L1-penalized estimation. *Statistics and Computing*, 24(2), 137–154. <https://doi.org/10.1007/s11222-012-9359-z>
- Hackett, G., & Betz, N. E. (1989). An exploration of the mathematics self-efficacy/mathematics performance correspondence. *Journal for Research in Mathematics Education*, 20(3), 261–273. <https://doi.org/10.2307/749515>
- Han, J., Kelley, T., & Knowles, J. G. (2021). Factors influencing student STEM learning: Self-efficacy and outcome expectancy, 21st century skills, and career awareness. *Journal for STEM Education Research*, 4(2), 117–137. <https://doi.org/10.1007/s41979-021-00053-3>
- Hox, J. J., Moerbeek, M., & van de Schoot, R. (2017). *Multilevel analysis: Techniques and applications* (3rd ed.). Routledge.
- Huang, Y., Zhou, Y., Chen, J., & Wu, D. (2024). Applying Machine Learning and SHAP Method to Identify Key Influences on Middle-School Students' Mathematics Literacy Performance. *Journal of Intelligence*, 12(10), 93. <https://doi.org/10.3390/jintelligence12100093>
- Kim, T. (2009). Confucianism, modernities and knowledge: China, South Korea and Japan. In R. Cowen & A. M. Kazamias (Eds.), *International handbook of comparative education* (pp. 857–872). Springer.
- Kong, S. C., Chan, T. W., Huang, R., & Cheah, H. M. (2014). A review of e-Learning policy in school education in Singapore, Hong Kong, Taiwan, and Beijing: implications to future policy planning. *Journal of Computers in Education*, 1(2), 187–212. <https://doi.org/10.1007/s40692-014-0011-0>
- Kriegbaum, K., Jansen, M., & Spinath, B. (2015). Motivation: A predictor of PISA's mathematical competence beyond intelligence and prior test achievement. *Learning and Individual Differences*, 43, 140–148. <https://doi.org/10.1016/j.lindif.2015.08.026>
- Lee, J. (2009). Universals and specifics of math self-concept, math self-efficacy, and math anxiety across 41 PISA 2003 participating countries. *Learning and Individual Differences*, 19(3), 355–365. <https://doi.org/10.1016/j.lindif.2008.10.009>
- Lee, J., & Stankov, L. (2018). Non-cognitive predictors of academic achievement: Evidence from TIMSS and PISA. *Learning and Individual Differences*, 65, 50–64. <https://doi.org/10.1016/j.lindif.2018.05.009>
- Lee, J., Lee, H. J., & Bong, M. (2022). Boosting children's math self-efficacy by enriching their growth mindsets and gender-fair beliefs. *Theory into Practice*, 61(1), 35–48. <https://doi.org/10.1080/00405841.2021.1932156>
- Lee, W., Lee, M.-J., & Bong, M. (2014). Testing interest and self-efficacy as predictors of academic self-regulation and achievement. *Contemporary Educational Psychology*, 39(2), 86–99. <https://doi.org/10.1016/j.cedpsych.2014.02.002>
- Lent, R. W., Brown, S. D., & Hackett, G. (1994). Toward a unifying social cognitive theory of career and academic interest, choice, and performance. *Journal of Vocational Behavior*, 45(1), 79–122. <https://doi.org/10.1006/jvbe.1994.1027>
- Lent, R. W., Lopez, F. G., & Bieschke, K. J. (1991). Mathematics self-efficacy: Sources and relation to science-based career choice. *Journal of Counseling Psychology*, 38(4), 424–430. <https://doi.org/10.1037/0022-0167.38.4.424>

- Lin, S. H., Lu, W. C., Chen, M. Y., & Chen, L. H. (2014). Association between proactive personality and academic self-efficacy. *Current Psychology*, 33(4), 600–609. <https://doi.org/10.1007/s12144-014-9231-8>
- Lloyd, J. E., Walsh, J., & Yailagh, M. S. (2005). Sex differences in performance attributions, self-efficacy, and achievement in mathematics: if I'm so smart, why don't I know it? *Canadian Journal of Education*, 28(3), 384–408. <https://doi.org/10.2307/4126476>
- Lopez, F. G., Lent, R. W., Brown, S. D., & Gore, P. A. (1997). Role of social-cognitive expectations in high school students' mathematics-related interest and performance. *Journal of Counseling Psychology*, 44(1), 44–52. <https://doi.org/10.1037/0022-0167.44.1.44>
- Meinshausen, N., & Bühlmann, P. (2010). Stability selection. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(4), 417–473. <https://doi.org/10.1111/j.1467-9868.2010.00740.x>
- Niu, J., Xu, H., & Yu, J. (2025). Identifying multilevel factors on student mathematics performance for Singapore, Korea, Finland, and Denmark in PISA 2022: Considering individualistic versus collectivistic cultures. *Humanities and Social Sciences Communications*, 12(1), 151. <https://doi.org/10.1057/s41599-025-04466-y>
- OECD. (2013). *PISA 2012 Results: Ready to Learn: Students' Engagement, Drive and Self-Beliefs (Volume III)*, PISA. OECD Publishing. <https://doi.org/10.1787/9789264201170-en>
- OECD. (2023). *PISA 2022 Assessment and Analytical Framework*, PISA: OECD Publishing. <https://doi.org/10.1787/dfe0bf9c-en>
- OECD. (2024a). *PISA 2022 results (Volume III): Creative minds, creative schools*. OECD Publishing. <https://doi.org/10.1787/765ee8c2-en>
- OECD. (2024b). *PISA 2022 results (Volume V): Learning strategies and attitudes for life*. OECD Publishing. <https://doi.org/10.1787/c2e44201-en>
- OECD. (2024c). *PISA 2022 technical report*. OECD Publishing. <https://doi.org/10.1787/01820d6d-en>
- Özcan, Z. Ç., & Eren Gümüş, A. (2019). A modeling study to explain mathematical problem-solving performance through metacognition, self-efficacy, motivation, and anxiety. *Australian Journal of Education*, 63(1), 116–134. <https://doi.org/10.1177/0004944119840073>
- Pajares, F. (1996). Self-efficacy beliefs in academic settings. *Review of Educational Research*, 66(4), 543–578. <https://doi.org/10.3102/00346543066004543>
- Pajares, F., & Graham, L. (1999). Self-efficacy, motivation constructs, and mathematics performance of entering middle school students. *Contemporary Educational Psychology*, 24(2), 124–139. <https://doi.org/10.1006/ceps.1998.0991>
- Pajares, F., & Miller, M. D. (1994). Role of self-efficacy and self-concept beliefs in mathematical problem solving: A path analysis. *Journal of Educational Psychology*, 86(2), 193–203. <https://doi.org/10.1037/0022-0663.86.2.193>
- Peters, M. L. (2013). Examining the relationships among classroom climate, self-efficacy, and achievement in undergraduate mathematics: A multi-level analysis. *International Journal of Science and Mathematics Education*, 11(2), 459–480. <https://doi.org/10.1007/s10763-012-9347-y>
- Pongsophon, P. (2025). Decoding Singapore's mathematics mastery: Student and school predictors of PISA 2022 performance. *Canadian Journal of Science, Mathematics and Technology Education*, 25(2), 308–323. <https://doi.org/10.1007/s42330-025-00349-1>
- Powell, A., & Barbour, M. K. (2018). A case study of e-learning initiatives in Singapore's secondary schools. *Handbook of research on K-12 online and blended learning* (2nd ed., pp. 631–648).
- Recher, S., Isiksal, M., & Koç, Y. (2018). Investigating self-efficacy, anxiety, attitudes and mathematics achievement regarding gender and school type. *Anales de Psicología/Annals of Psychology*, 34(1), 41–51. <https://doi.org/10.6018/analesps.34.1.229571>
- Rosenthal, H. E., Crisp, R. J., & Suen, M. W. (2007). Improving performance expectancies in stereotypic domains: Task relevance and the reduction of stereotype threat. *European Journal of Social Psychology*, 37(3), 586–597. <https://doi.org/10.1002/ejsp.379>
- Saenz, M. B., Nandakumar, V., & Adamuti-Trache, M. (2023). A Comparative Study of High School Students' Math Achievement and Attitudes: Do Math Teacher Qualifications Matter? *International Journal of Education in Mathematics, Science and Technology*, 11(2), 304–322. <https://doi.org/10.46328/ijemst.2528>
- Schunk, D. H. (1991). Self-efficacy and academic motivation. *Educational Psychologist*, 26(3-4), 207–231. https://doi.org/10.1207/s15326985ep2603&4_2
- Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464. <https://doi.org/10.1214/aos/1176344136>
- Shavelson, R. J., Hubner, J. J., & Stanton, G. C. (1976). Self-concept: Validation of construct interpretations. *Review of Educational Research*, 46(3), 407–441. <https://doi.org/10.3102/00346543046003407>
- Shevade, S. K., & Keerthi, S. S. (2003). A simple and efficient algorithm for gene selection using sparse logistic regression. *Bioinformatics*, 19(17), 2246–2253. <https://doi.org/10.1093/bioinformatics/btg308>
- Siegle, D., & McCoach, D. B. (2007). Increasing student mathematics self-efficacy through teacher training. *Journal of Advanced Academics*, 18(2), 278–312.

- Stankov, L. (2010). Unforgiving Confucian culture: A breeding ground for high academic achievement, test anxiety and self-doubt? *Learning and Individual Differences*, 20(6), 555–563. <https://doi.org/10.1016/j.lindif.2010.05.003>
- Steele, C. M. (1997). A threat in the air: How stereotypes shape intellectual identity and performance. *American Psychologist*, 52(6), 613–629. <https://doi.org/10.1037//0003-066x.52.6.613>
- Street, K. E. S., Malmberg, L.-E., & Schukajlow, S. (2024). Students' mathematics self-efficacy: a scoping review. *ZDM—Mathematics Education*, 56(2), 265–280. <https://doi.org/10.1007/s11858-024-01548-0>
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1), 267–288.
- Usher, E. L., & Pajares, F. (2009). Sources of self-efficacy in mathematics: A validation study. *Contemporary Educational Psychology*, 34(1), 89–101. <https://doi.org/10.1016/j.cedpsych.2008.09.002>
- Usher, E. L., Li, C. R., Butz, A. R., & Rojas, J. P. (2019). Perseverant grit and self-efficacy: Are both essential for children's academic success? *Journal of Educational Psychology*, 111(5), 877–902. <https://doi.org/10.1037/edu0000324>
- Usta, H. G. (2016). Analysis of Student and School Level Variables Related to Mathematics Self-Efficacy Level Based on PISA 2012 Results for China-Shanghai, Turkey, and Greece. *Educational sciences: Theory and Practice*, 16(4), 1297–1323.
- Wang, Y., Wang, Y., Pan, Z., & Ortega-Martín, J. L. (2024). The predicting role of EFL students' achievement emotions and technological self-efficacy in their technology acceptance. *The Asia-Pacific Education Researcher*, 33(4), 771–782. <https://doi.org/10.1007/s40299-023-00750-0>
- Wong, Z. Y., & Liem, G. A. D. (2025). School-Based Career Guidance in Singapore: Implications on Identity, Social, and Emotional Development. In *In Social and Emotional Development for Diverse Learners: The Addition of Asian Perspectives* (pp. 211–229): Springer Nature.
- Wu, Y. (2016). Universal Beliefs and Specific Practices: Students' Math Self-Efficacy and Related Factors in the United States and China. *International Education Studies*, 9(12), 61–74. <https://doi.org/10.5539/ies.v9n12p61>
- Xiang, X., Yum, S., & Kang, D. (2015). Factors Affecting on Student Math Self-efficacy of Korea and Singapore based on PISA 2012. *The Mathematical Education*, 54(1), 49–63. <https://doi.org/10.6331/mathedu.2015.54.1.49>
- Yoo, J. E. (2024). Penalized Regression in Large-Scale Data Analysis. In M. S. Khine (Ed.), *Machine Learning in Educational Sciences: Approaches, Applications and Advances* (pp. 71–91). Springer.
- Yoo, J. E., & Rho, M. (2020). Exploration of predictors for Korean teacher job satisfaction via a machine learning technique, Group Mnet. *Frontiers in Psychology*, 11, 441. <https://doi.org/10.3389/fpsyg.2020.00441>
- Yoo, J. E., & Rho, M. (2022). Large-scale survey data analysis with penalized regression: A Monte Carlo simulation on missing categorical predictors. *Multivariate Behavioral Research*, 57(4), 642–657. <https://doi.org/10.1080/00273171.2021.1891856>
- Yurt, E. (2014). The predictive power of self-efficacy sources for mathematics achievement. *Education and Science*, 39(176), 159–169. <https://doi.org/10.15390/eb.2014.3443>
- Zakariya, Y. F. (2022). Improving students' mathematics self-efficacy: A systematic review of intervention studies. *Frontiers in Psychology*, 13, 986622. <https://doi.org/10.3389/fpsyg.2022.986622>
- Zientek, L. R., Fong, C. J., & Phelps, J. M. (2019). Sources of self-efficacy of community college students enrolled in developmental mathematics. *Journal of Further and Higher Education*, 43(2), 183–200. <https://doi.org/10.1080/0309877x.2017.1357071>

Appendix

Table 5. Summary of PISA 2022 Student- and School-Level Constructs From the Core Context Questionnaires

Level	Framework	Constructs
Student-level (16/36)	Student Background (0/6)	Gender (C), Number of siblings (C), Index of economic, social, and cultural status (CC), Time attended current school, Grade repetition (SI/C), Family support (SC)
	Student Beliefs, Attitudes, Feelings, and Behaviors (9/26)	Mathematics teacher support (M/SC), Mathematics motivation (M/SI/C), Mathematics self-concept (M/SI/C), Mathematics preference (M/SI/C), Growth mindset (M/SC), Mathematics anxiety (M/SC), Proactive mathematics study behavior (M/SC), Time spent on mathematics homework (M), Participation in additional mathematics instruction (M/sum) Missing school (SI/C), Skipping classes or days of school (SI/C), Arriving late for school (SI/C), Highest expected educational level (SI), Clear idea about future job (SI/C), Being bullied (SC), Feeling safe (SC), Quality of student–teacher relationships (SC), School safety risks (SC), Sense of belonging (SC), Assertiveness (SC), Cooperation (SC), Curiosity (SC), Emotional control (SC), Empathy (SC), Perseverance (SC), Stress resistance (SC), Time spent on online activities (sum), Exercising or practicing a sport before or after school (SI), Studying for school or homework before or after school (SI), Working for pay before or after school (SI), Working in household or taking care of family members (SI), Beliefs about creativity (sum), Creative thinking self-efficacy (SC), Creativity and openness to intellect (SC), Openness to art and reflection (SC)
	Teaching Practices and Learning Opportunities (7/4)	Subjective familiarity with mathematics concepts (M/SC), Class periods per week in mathematics (M), Exposure to formal and applied mathematics tasks (M/SC), Exposure to mathematical reasoning and 21st-century mathematics tasks (M/SC), Disciplinary climate in mathematics (M/SC), Cognitive activation in mathematics: Foster reasoning (M/SC), Encourage mathematical thinking (M/SC) Creative school and class environment (SC), Creative peers and family environment (SC), Participation in creative activities at school (SC), creative activities outside of school (SC)

(continued)

Table 5. (continued)

Level	Framework	Constructs
School-level (13/43)	Student Beliefs, Attitudes, Feelings, and Behaviors (0/1)	Openness culture/climate (SC)
	Teaching Practices and Learning Opportunities (5/3)	Math class size (M/SI), Average time in class period (M), Mathematics teacher qualifications: Full time (M), Part time (M), Mathematics teacher training (M/SC) In-house professional development (sum), Proportion of all teachers fully certified (SI), Creative school environment (SC)
	School Practices, Policies, and Infrastructure (6/30)	Mathematics-related extra-curricular activities at school (M/SI), Proportion of mathematics teachers at school (M/SI), Student-mathematics teacher ratio (M/SI), Ability grouping for mathematics class (M/SI/C), Percent of students at/above pass mark in mathematics (M), Standardized mathematics curriculum (M/sum) Career guidance (C), Information about future educational/work/career path (sum), Percentage of students with migration background, Number of instructional languages, Factors hindering instruction (sum), Student-related factors affecting school climate (SC), Teacher-related factors affecting school climate (SC), Social and emotional learning policies (C), Extra-curricular activities offered (SC), Creative extracurricular activities offered (SI), School type (SI/C), Total number of students, Availability of computers (SI), Availability of tablet devices (SI), Computers connected to internet (SI), Proportion of personnel for pedagogical support (SI), school administrative personnel (SI), management personnel (SI), Total number of nonteaching staff at school (SI), Proportion of all teachers with at least ISCED level 7 master qualification (CI), Proportion of all teachers with ISCED level 8 doctoral qualification (CI), Student-teacher ratio (SI), School selection competition (C), Academic school selectivity (SI/C), School offering study help (sum), Student ability grouping/tracking policies and practices (sum), Monitoring teacher practices (sum), Feedback to teachers (SC), Creative school activities offered (SC), Beliefs about creativity (SC)
	Governance, System-level Policies and Practices (2/9)	Use of mathematics assessments (M/C), Use of mathematics achievement data in accountability systems (M/sum) School autonomy (SC), Teacher participation (SC), School responsibility for curriculum (SI), School responsibility for resources (SI), Use of standardized tests (SC), Use of teacher-developed tests (SC), School evaluation (sum), School encouragement of parent or guardian involvement (SC), Parent/guardian participation in activities (sum)

Note. C: categorical variable; CC: complex composite indices; M: mathematics-specific constructs; SC: scaled derived variable; SI: simple derived variable; sum: manually summed or mean averaged by researchers. Variables that are not categorical are considered continuous. Categorization is based on the PISA 2022 Analytical Framework (OECD, 2023).